ALICE L WALTON SCHOOL OF MEDICINE

# Antimetabolite Agents

Last updated April 22, 2025 10 min read ScholarRx

**Adapted by:** Trager Hintze, Chelsey Deel, Diptiman Bose

**Editors:** Russell Johnson, Jordan Finn

Adapted from: Antimetabolite Agents

Listen to this Brick
0:00 / 19:20 1x

Learning Objectives (3) Versions

*After completing this brick, you will be able to:*

1. Classify antimetabolite agents by the three groups: antifolates, purine nucleoside analogs, and pyrimidine nucleoside analogs.
2. Explain the mechanism of action of antimetabolite agents.
3. Explain the adverse effects of antimetabolite agents and discuss possible strategies for minimizing/preventing these effects.

**CASE CONNECTION**

JR is a 61-year-old man who presents with persistent fatigue, night sweats, and unintentional weight loss over the past several months. A lymph node biopsy confirms a diagnosis of B-cell non-Hodgkin lymphoma. After staging and discussion of treatment options, you recommend cladribine, a purine analog chemotherapy agent known for its effectiveness in treating indolent B-cell malignancies.

"I read that cladribine suppresses the immune system," JR says. "I'm already feeling weak, and I'm worried about infections. Is this really the best option for me?"

How will you explain to JR how cladribine works, and how it could be effective in treating his lymphoma? Consider your answer as you read, and we'll revisit at the end of the brick.

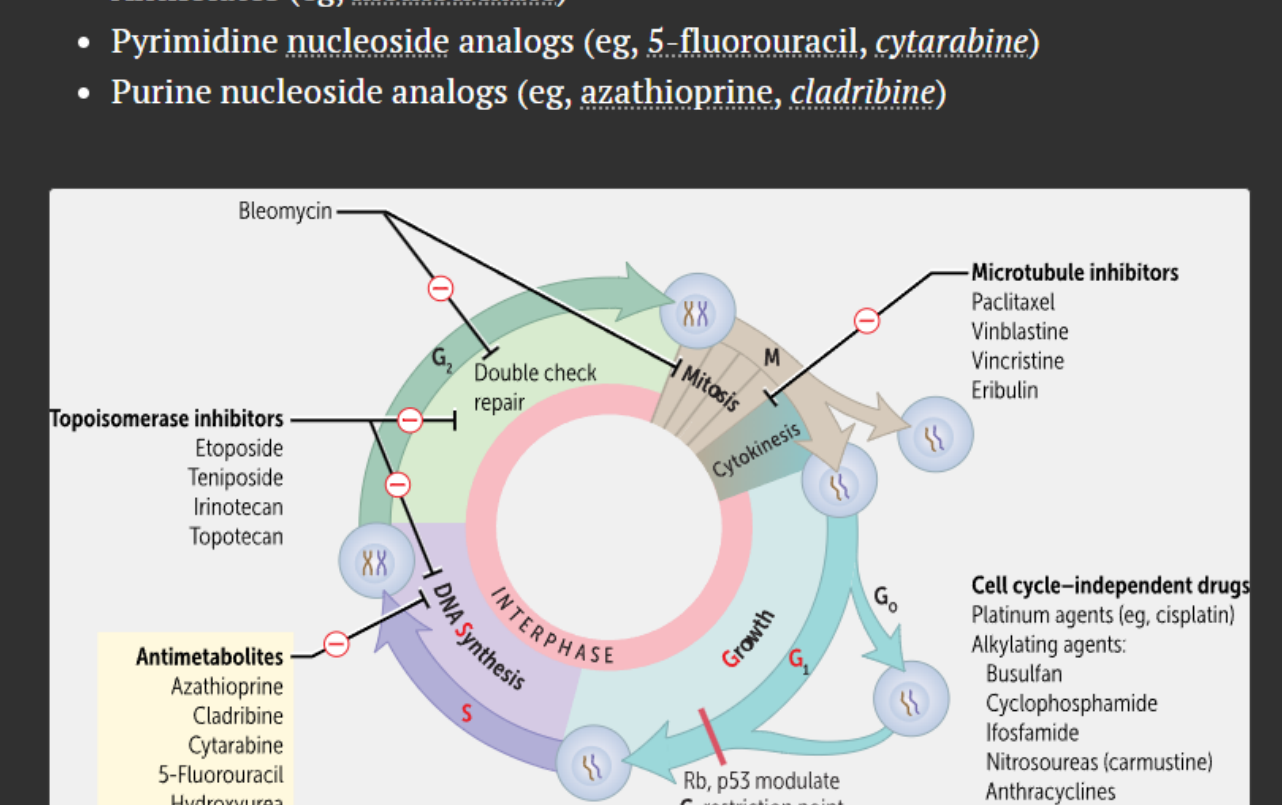
GO TO CONCLUSION ↓

## What Are Antimetabolite Agents? (LO#1)

Cancer cells replicate their DNA in an unregulated manner, incorporating nucleosides into their growing DNA chains. In their ambitiousness, cancer cells incorporate compounds that resemble the nucleosides. This point of weakness is exploited by chemotherapy with antimetabolites. These drugs are structural analogs of different substrates used in the normal synthesis of pyrimidine and purine nitrogenous bases. They therefore act as decoys, interrupting normal DNA synthesis and causing the cell to die.

The antimetabolite class of drugs was the first to be successfully used in the treatment of cancer. These agents inhibit DNA synthesis during the DNA synthesis phase (S phase) of the cell cycle (Figure 1). They include:

- Antifolates (eg, methotrexate)
- Pyrimidine nucleoside analogs (eg, 5-fluorouracil, *cytarabine*)
- Purine nucleoside analogs (eg, azathioprine, *cladribine*)



QUIZ

Figure 1

## What Are Antifolate Agents? (LO#2)

Antifolates inhibit the metabolism of **folic acid** (vitamin B9), a water-soluble vitamin obtained from folate-enriched grains (breads) as well as vegetables (broccoli, spinach, beans, greens) and liver. Folic acid is necessary for the formation of the DNA nucleotide **thymine**. The antifolates inhibit the pathway of thymine formation, so normal DNA cannot form. Rapidly growing cells like cancer cells then undergo apoptosis, or programmed cell death. So just how does folate work in thymine synthesis? It uses a particular biochemical pathway.

### One-Carbon Metabolism

Folate metabolism supports the broader set of biochemical processes known as one-carbon (1C) metabolism. This universal metabolic process activates and transfers 1C units for biosynthetic processes including purine and thymidine. Whether in a healthy cell or a cancer cell, the process of nucleotide synthesis involves the transfer of single carbon atoms. An example of a 1C pathway relevant to cancer is the synthesis of thymine (dTMP), which requires the addition of a single carbon atom to uracil (dUMP) using the enzyme thymidylate synthase as well as methylated tetrahydrofolate (THF), a cofactor (Figure 2). Methylated THF is produced from folic acid, first via sequential reduction to dihydrofolate (DHF) and tetrahydrofolate (THF), then via methylation.

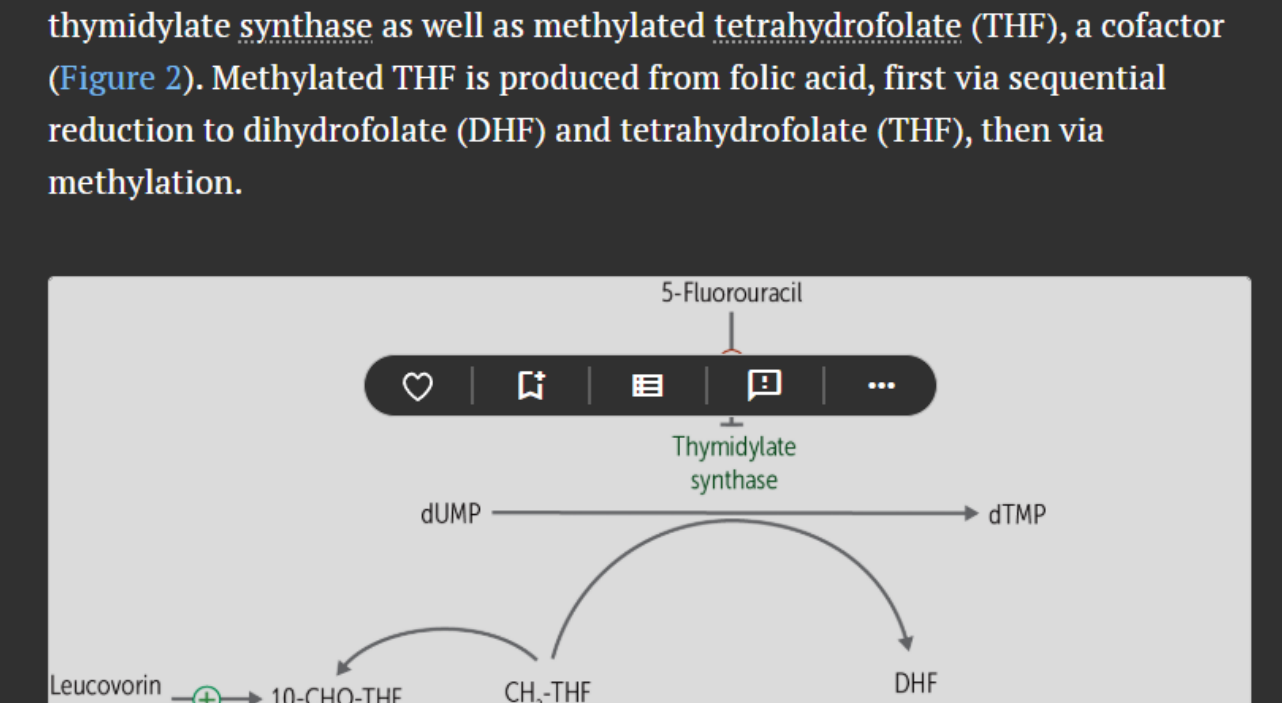


Figure 2

### Mechanism of Action of Antifolates

Because actively dividing cancer cells use lots of thymidine, inhibition of this dTMP synthesis pathway effectively kills cancer cells. This is where antifolate chemotherapeutic agents such as **methotrexate** (**MTX**) enter the scene. Methotrexate inhibits dihydrofolate reductase (see Figure 2). By targeting one-carbon metabolism, antimetabolites interfere with the replication of DNA in cancer cells. They are particularly effective with rapid cell proliferation.

**CLINICAL CORRELATION**

Folate is a nutrient especially important for DNA production in cells with rapid turnover such as hair, nails, and red blood cells (RBCs). The body maintains stores of folate that can las[illegible]ancy, in addition to several other conditions, accelerates the depletion of folate stores. During pregnancy, the growing fetus makes use of the mother's folate stores to synthesize its DNA. This is why pregnant patients are advised to take folic acid supplements. A deficiency of folate in pregnancy can lead to neural tube defects.

How does methotrexate inhibit cancer cell proliferation?

### Clinical Uses of Methotrexate

Methotrexate is the most commonly used antimetabolite agent and is prescribed to treat leukemias, lymphomas, breast cancer, choriocarcinomas, and sarcomas.

**CLINICAL CORREL[illegible]**

Methotrexate is also used for its immunosuppressive activity in autoimmune disorders such as rheumatoid arthritis, psoriasis, Crohn disease, and systemic lupus erythematosus. Autoimmune disorders occur when the immune system targets the healthy cells in the body. In such cases, MTX is used to suppress the production of immune cells, including autoimmune cells.

### Adverse Effects of Methotrexate (LO#3)

Methotrexate has a high affinity for the rapidly dividing cells produced in the bone marrow. Therefore, myelosuppression (decreased bone marrow blood cell formation) is a common complication. A patient with MTX-induced myelosuppression may present with pancytopenia (reductions in all blood cell types). In **megaloblastic anemia**, a reduction in RBCs with enlargement (high mean corpuscular volume) and large, immature megaloblasts in the bone marrow are seen. A reduction in platelets leads to easy bleeding or bruising. A reduction in leukocytes causes frequent infections.

**Leucovorin Rescue.** MTX-induced myelosuppression can be reversed with leucovorin, a folic acid analog that does not require activation by DHF reductase, so it bypasses the blocked pathway and "rescues" the formation of thymidine.

Other adverse effects of methotrexate include pulmonary fibrosis, liver toxicity, and inflamed mucous membranes in the mouth and stomach (mucositis) leading to a sore mouth, nausea, and vomiting. MTX is also a pregnancy category X drug because it can cause neural tube defects in the fetus.

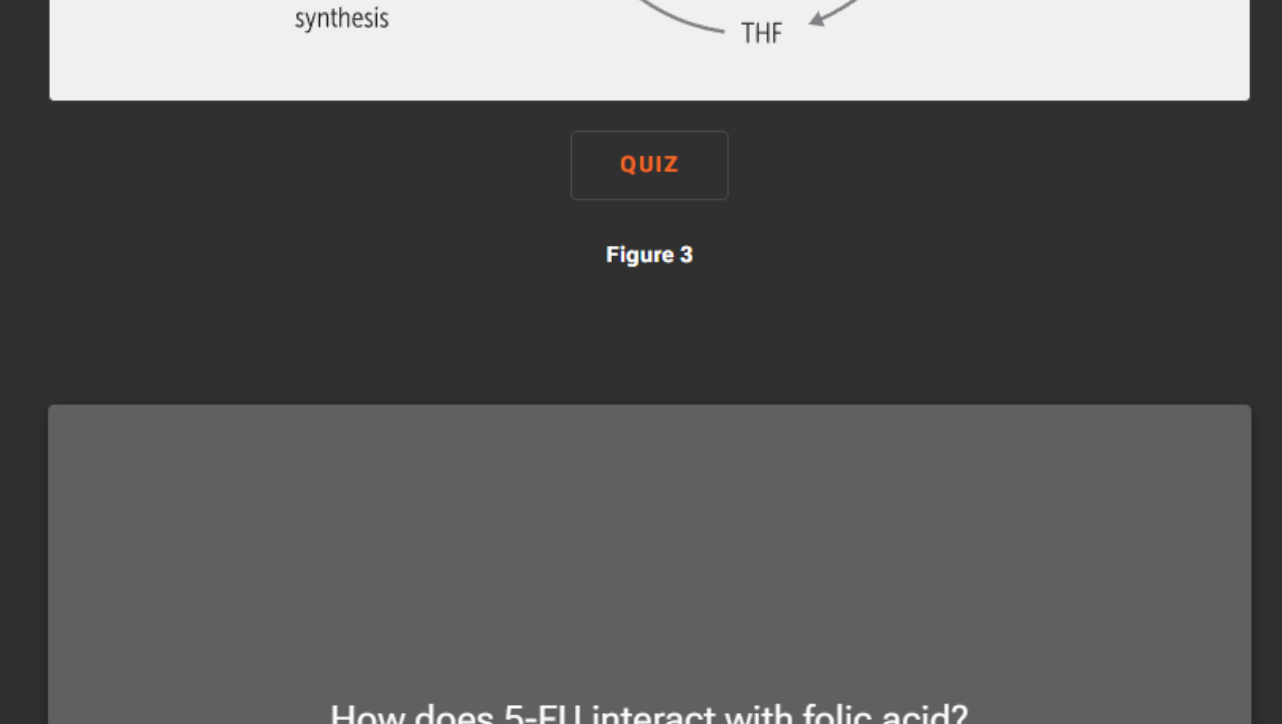
Why is leucovorin given to patients receiving methotrexate therapy?

## What Are Pyrimidine Nucleoside Analogs? (LO#2)

Drugs such as 5-fluorouracil, **cytarabine**, and capecitabines are structurally similar to purine or pyrimidine bases that form DNA.

### Mechanism of Action of Pyrimidine Analogs

**5-Fluorouracil** (**5-FU**) is the most studied and commonly used nucleoside analog. After being metabolized into active metabolites, 5-FU binds and forms a complex with folic acid. This complex binds to thymidylate synthase, rendering it dysfunctional (Figure 3). Without thymidylate synthase, the single carbon atom cannot be transferred to dUMP to produce dTMP.



QUIZ

Figure 3

How does 5-FU interact with folic acid?

### Clinical Uses of Pyrimidine Analogs

5-FU prodrugs are inactive when ingested but are then converted into 5-FU within the tumor. **Cytarabine**, for example, is converted to 5-FU by the enzyme thymidine phosphorylase, which has greater expression in tumors than in healthy tissues, potentially reducing adverse effects.

Cytarabine is used to treat some leukemias and lymphomas.

In contrast to its reversal of the action of MTX, **leucovorin** enhances the action of 5-FU. This is because 5-FU forms a more stable complex with leucovorin than it does with folic acid. Because of this additive effect, leucovorin is often combined with 5-FU to increase its chemotherapeutic potency.

### Adverse Effects of Pyrimidine Analogs (LO#3)

Common adverse effects of 5-FU include nausea and vomiting, diarrhea, mucositis, and skin rashes and redness, especially on the extremities (hand-foot syndrome). 5-FU may cause myelosuppression; gastrointestinal toxicity such as diarrhea, nausea, and vomiting; hepatotoxicity; and pulmonary fibrosis. Because leucovorin augments the effects of 5-FU, the myelosuppression can be severe in patients receiving both 5-FU and leucovorin.

What is a common severe side effect of combining 5-FU with leucovorin?

## What Are Purine Nucleoside Analogs? (LO#2)

Drugs such as azathioprine, 6-mercaptopurine, and **cladribine** are structurally similar to purine bases that form DNA.

### Mechanism of Action of Purine Analogs

**Cladribine** is a purine nucleoside analogue. It is a prodrug which is activated by phosphorylation and converted into the active form, Cd-ATP. This active form incorporates into DNA to result in the breakage of DNA strand and shutdown of DNA synthesis and repair. This also results in a depletion of nicotinamide adenine dinucleotide and adenosine triphosphate (ATP).

What is the active form of cladribine?

### Clinical Uses of Purine Analogs

Cladribine is used in some lymphomas and has recently been approved for use in multiple sclerosis.

**CLINICAL CORRELATION**

Azathioprine is the most common of the purine analogs, however, it is used mostly for autoimmune diseases such as rheumatoid arthritis, systemic lupus erythematosus, vasculitis, and inflammatory bowel disease. It is also used in some anti-rejection regimens after organ transplantation.

### Adverse Effects of Purine Analogs (LO#3)

Similar to the pyrimi[illegible]uses nausea, vomiting, and myelosuppression.

The antimetabolites are summarized in Table 1.

Table 1

| Drug | Class | Mechanism | Major Adverse Effects |
|---|---|---|---|
| Azathioprine<br>6-Mercaptopurine | Purine analog | Incorporates into DNA during replication and induces apoptosis | Myelosuppression, hepatotoxicity, increased toxicity with allopurinol |
| Cladribine | Purine analog | Incorporates into DNA and prevents transcription; impairs DNA repair | Myelosuppression, nephrotoxicity, neurotoxicity |
| Cytarabine | Pyrimidine analog | Incorporates into DNA, causes early chain termination; inhibits DNA polymerase | Pancytopenia |
| 5-Fluorouracil | Pyrimidine analog | Inhibits thymidylate synthase | Myelosuppression (worsens with leucovorin) |
| Methotrexate | Folic acid analog | Inhibits dihydrofolate reductase | Myelosuppression (reversible with leucovorin), hepatotoxicity, pulmonary fibrosis |

**CASE CONNECTIO[illegible]** [illegible]K TO INTRODUCTION ↑

Thinking back to JR, what do you tell him about the treatment recommendations?

You explain to JR that cladribine is an effective treatment for B-cell non-Hodgkin lymphoma, particularly in indolent forms of the disease, and that early intervention can help control the cancer and improve long-term outcomes. You acknowledge his concerns about immunosuppression and reassure him that his blood counts will be closely monitored throughout treatment, with regular follow-ups to check for infections and any potential side effects.

"I'll also be watching your kidney and liver function to ensure your body is tolerating the medication well," you add. "We'll take steps to reduce infection risks, such as considering prophylactic antibiotics if needed."

"That's good to know," JR says. "I want to do everything I can to stay strong during treatment."

## Summary

- Antimetabolites are decoys of metabolic intermediates in DNA synthesis; major subclasses include antifolates and nucleoside analogs.
- Antimetabolites target intermediates in one-carbon metabolism; the incorporation of an antimetabolite terminates DNA synthesis in cancer cells.
- Folate is an important cofactor in one-carbon metabolism and the production of the pyrimidine base, thymine; methotrexate and 5-fluoruracil both inhibit the production of thymine.
- Methotrexate is a folic acid analog that inhibits dihydrofolate reductase.
- An adverse effect[illegible]; it can be reversed with leucovorin rescue.
- The nucleoside analog subclass of antimetabolites includes the purine and pyrimidine analogs.
- Cytarabine is a prodrug that is converted to 5-Fluorouracil, a pyrimidine analog, that forms a complex with folic acid and inhibits thymidylate synthase.
- Myelosuppression is also an adverse event of 5-fluorouracil.
- Cladribine is a purine nucleoside analog that substitutes into the DNA chain and blocks replication. It is used in some lymphomas and has recently been approved for use in multiple sclerosis.

## Review Questions

**1. Antimetabolites target which phase of the cell cycle?**

A. ○ Cell-cycle independent
B. ○ $G_1$
C. ○ $G_2$
D. ○ M
E. ● S

**Hide Explanation**

The correct answer is [illegible]sis, which only occurs during the synthesis phase of the cell cycle. Antimetabolites are therefore not cell-cycle independent (A), which is a feature of the alkylating agents. $G_1$ (B), $G_2$ (C), and M (D) phases are less characterized by DNA synthesis and therefore less likely to be affected by the antimetabolite class of agents.

**2. What is the active form of cladribine?**

A. ○ Adenosine triphosphate
B. ● Cd-ATP
C. ○ Cd-AMP
D. ○ Cd-ADP
E. ○ Cladribine-TP

**Hide Explanation**

The correct answer is Cd-ATP (B). Cladribine is converted into Cd-ATP, which incorporates into DNA and disrupts its synthesis and repair. Adenosine triphosphate (A) is a general energy carrier in cells, not specific to cladribine. Cd-AMP (C) and Cd-ADP (D) are not the active forms of cladribine. Cladribine-TP (E) is not a recognized form.

**3. What is the most common adverse event associated with methotrexate use?**

A. ○ Cardiotoxicity
B. ○ Gastrointestinal toxicity
C. ● Myelosuppression
D. ○ Ototoxicity
E. ○ Peripheral neuropathy

**Hide Explanation**

The correct answer is myelosuppression (C), a feared complication with most chemotherapy drugs that occurs even more commonly with methotrexate chemotherapy. Doxorubicin, a DNA intercalator, is known for its cardiotoxicity (A). 5-Fluorouracil is known for its gastrointestinal toxicity (B), because it is heavily metabolized in the intestines. Platinum-based chemotherapy drugs are known for ototoxicity (D). Vincristine, a microtubule inhibitor, causes peripheral neuropathy (E).

**How would you rate this Brick?**

chemotherapy drugs that occurs even more commonly with methotrexate chemotherapy. Doxorubicin, a DNA intercalator, is known for its cardiotoxicity (A). 5-Fluorouracil is known for its gastrointestinal toxicity (B), because it is heavily metabolized in the intestines. Platinum-based chemotherapy drugs are known for ototoxicity (D). Vincristine, a microtubule inhibitor, causes peripheral neuropathy (E).

How would you rate this Brick?

## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Classify antimetabolite agents by the three groups: antifolates, purine nucleoside analogs, and pyrimidine nucleoside analogs.~~
02 ☑ ~~Explain the mechanism of action of antimetabolite agents.~~
03 ☑ ~~Explain the adverse effects of antimetabolite agents and discuss possible strategies for minimizing/preventing these effects.~~

**Objective Confidence: 3/3**